

Name: _____



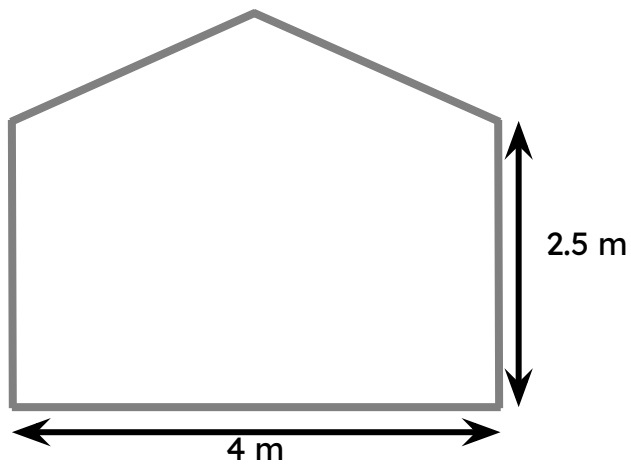
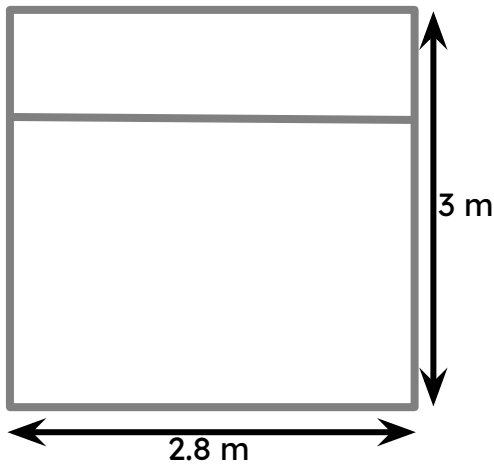
Construction

Task A: Using formulae in the construction industry

1) Here are the front and side elevations of a shed.

A builder needs to replace the rafters of the shed.

How long are the rafters?



2) $Q = A \times v$ where Q is the flow rate, A is the flow area and v is the velocity

Water runs through a cylindrical pipe at a velocity of 10 m/s.

The diameter of the pipe is 40 mm.

What is the flow rate in m^3 per hour?

3) $P = I \times V$ where P is the power (watts), I is the current (amps) and V is the voltage (volts)

The voltage of electricity in UK households is 230 volts.

An electrician needs to install a 2.15 kW electric oven.

Circuits are available in 6, 10, 16, 20, 32, 40, 50 and 63 amps.

Which circuit should be installed?

4) BTU = volume of the room $\times$ allowance for heat loss

Below are the dimensions of a bedroom.

Length = 4.6 m

Width = 2.6 m

Height = 2.4 m

a) Calculate the BTU.

Room	Allowance
Bathroom	121.5
Bedroom	108
Dining room	135
Hallway	81
Kitchen	81
Living room	135

b) The bedroom will have one radiator.

Which size should be installed?

Height (mm)	Width (mm)	BTU
600	400	2184
600	600	3280
600	800	4372
600	1000	5464
600	1200	6556
600	1400	7649
600	1600	8744

Task B: Costing in the construction industry

1) This table shows the prices and lengths wood is sold in.

Length (mm)	Price
2400	£4.20
3000	£5.25
3600	£6.30
4200	£7.35
4800	£8.40

Calculate the cheapest price for purchasing the following:

a) Twenty 1 m lengths

b) Fifteen 1.8 m lengths

c) 3 different ways to buy 600 1.2 m lengths

d) Why are there 3 different ways for part (c)?